

Igniters Innovation

Challenge 2026

IDEAS TO IMPACT



CASE STUDY-E TRACK >>>

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HYDROGEN PASSENGER CARS IN INDIA: FCEV VS H2-ICE VS BEV

Government Policy Context

India launched the National Green Hydrogen Mission (NGHM) in January 2023 with an outlay of ₹19,744 crore. The mission aims to make India a global hub for the production, utilization, and export of green hydrogen, targeting 5 million metric tonnes (MMT) of annual green hydrogen production by 2030. The initiative supports hydrogen adoption across industries, mobility, energy storage, and infrastructure development to reduce fossil fuel imports, lower carbon emissions, and achieve India's Net Zero target by 2070.

Challenge

Choose one of the following:

Option 1: Hydrogen Vehicle Concept

Develop a hydrogen-powered passenger vehicle concept (FCEV or H2-ICE) and compare it with BEVs in terms of:

- Total Cost of Ownership (TCO)
- Performance and range
- Safety
- Sustainability
- Infrastructure requirements

Option 2: Hydrogen Retrofit Solution

Evaluate the feasibility of converting existing gasoline, CNG, or LNG passenger vehicles to hydrogen at an affordable cost and compare with BEVs in terms of:

- Conversion cost and TCO
- Performance and range
- Safety
- Sustainability
- Infrastructure requirements

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Expected Outcome

Develop a practical strategy that answers:

Can hydrogen-powered passenger vehicles (FCEV or H2-ICE), including retrofit solutions, become a safe, affordable, and commercially viable complement to BEVs in India by 2030?

The solution should balance customer needs, technology, safety, and business feasibility.

Key Considerations

1. Market Potential & Customer Assessment

- Identify the target customer segments and key use cases for hydrogen-powered passenger vehicles in India.
- Review global hydrogen mobility trends and benchmark leading OEMs on their investments, technology strategies (FCEV and H2-ICE), and market adoption.
- Assess the potential applicability of these trends to the Indian market.

2. Technology Comparison

- Compare FCEV, H2-ICE, and BEV vehicle concepts technologies in terms of:
 - Range
 - Refueling/charging time
 - Vehicle cost
 - Energy efficiency
 - Infrastructure requirements
 - Sustainability

3. Safety Challenges & Solutions

- Hydrogen as a fuel – safety challenges compare to EV, LNG, CNG etc.
- Hydrogen storage safety
- Crashworthiness considerations
- Hydrogen leak detection and mitigation
- Fire Safety

4. Business Viability & Adoption Roadmap

- Hydrogen production, distribution, and refueling infrastructure
- Investment requirements and ecosystem readiness
- Recommendations for large-scale deployment in India

